

1 Lie Algebras and Representation Theory

Basic

Exercise 1.1. Let W_1 be the Witt algebra spanned by the basis $\{d_n = x^{n+1} \frac{d}{dx} \mid n \in \mathbb{Z}\}$ with the Lie bracket relations:

$$[d_n, d_m] = (m - n)d_{n+m}$$

1. Consider the subspace $\mathfrak{h} = \text{span}_{\mathbb{C}}\{d_{-1}, d_0, d_1\}$. By explicitly calculating the brackets $[d_0, d_{-1}]$, $[d_0, d_1]$, and $[d_1, d_{-1}]$, prove that $\mathfrak{h}$ is closed under the Lie bracket (i.e., it is a Lie subalgebra).
2. The standard basis $\{e, f, h\}$ of $\mathfrak{sl}(2, \mathbb{C})$ satisfies the relations $[h, e] = 2e$, $[h, f] = -2f$, and $[e, f] = h$. Give an explicit isomorphism of Lie algebras from $\mathfrak{sl}(2, \mathbb{C})$ to $\mathfrak{h}$.

Exercise 1.2. Let $A_1 = \langle x, y \mid yx - xy = 1 \rangle$ be the first Weyl algebra (over $\mathbb{C}$), $d = yx$.

1. Using the algebraic relations, rewrite x^2y^2 as a polynomial in d . That is, find constants a, b, c such that:

$$x^2y^2 = ad^2 + bd + c$$

2. Show that $\{x^i y^j \mid i, j \in \mathbb{N}\}$ is a basis of A_1 as a vector space over $\mathbb{C}$.

Intermediate

Exercise 1.3. Recall the definition of the modules $V(\alpha, \beta)$ for the Witt algebra W_1 . The underlying vector space is $z^\beta \mathbb{C}[z, z^{-1}](dz)^\alpha$ with the action:

$$d_n \cdot (z^\gamma (dz)^\alpha) = (\gamma + n\alpha) z^{\gamma+n} (dz)^\alpha$$

1. Check whether $V(-1, 0)$ is irreducible. If it is, prove it. If it is not, explicitly identify a non-trivial proper submodule.
2. Show that $V(1, 0)$ is reducible, describe its maximal proper submodule, denoted by W .
3. Prove that $W \simeq V(0, 0)/\mathbb{C} \cdot 1$.

Exercise 1.4. According to the Clebsch-Gordan decomposition theory, which is for simple $\mathfrak{sl}(2, \mathbb{C})$ modules $V(n)$ and $V(m)$ where $n \geq m$, $V(n) \otimes V(m) \simeq V(n+m) \oplus V(n+m-2) \oplus \dots \oplus V(n-m)$.

1. Decompose the tensor product of the irreducible $\mathfrak{sl}(2, \mathbb{C})$ -modules $V(3) \otimes V(1)$ into a direct sum of irreducible modules.
2. Let v_0 be the highest weight vector of $V(3)$ (weight 3) and u_0 be the highest weight vector of $V(1)$ (weight 1). In the decomposition above, the highest weight component (isomorphic to $V(4)$) is generated by $v_0 \otimes u_0$.

Using the action of the lowering operator f (via the rule $f(v \otimes u) = fv \otimes u + v \otimes fu$), find the specific linear combination of basis vectors that represents the **highest weight vector** of the component isomorphic to $V(2)$ (i.e., a vector of weight 2 that is annihilated by e).

Advanced

Exercise 1.5. Recall that the Casimir element in the universal enveloping algebra $U(\mathfrak{sl}(2, \mathbb{C}))$ is defined as $c = (h+1)^2 + 4fe$. It is known that c acts as a scalar $(\lambda+1)^2$ on any irreducible module with highest weight λ .

Consider the tensor product module $W = V(1) \otimes V(1) \otimes V(1)$ (note that the action of $x \in \mathfrak{sl}(2, \mathbb{C})$ on triple tensor is $x \cdot (v \otimes u \otimes w) = xv \otimes u \otimes w + v \otimes xu \otimes w + v \otimes u \otimes xw$). Let $\{v_1, v_{-1}\}$ be the standard weight basis for the first $V(1)$, $\{u_1, u_{-1}\}$ for the second $V(1)$, $\{w_1, w_{-1}\}$ for the third $V(1)$ (indices denote weights). The space W has a basis $\{v_1 \otimes u_1 \otimes w_1, v_1 \otimes u_1 \otimes w_{-1}, v_1 \otimes u_{-1} \otimes w_1, v_1 \otimes u_{-1} \otimes w_{-1}, v_{-1} \otimes u_1 \otimes w_1, v_{-1} \otimes u_1 \otimes w_{-1}, v_{-1} \otimes u_{-1} \otimes w_1, v_{-1} \otimes u_{-1} \otimes w_{-1}\}$.

1. Write down the decomposition of W into a direct sum of irreducible submodules.
2. How many highest weight vectors are there in W ? Write down explicit expressions for each of them in terms of the product basis above.
3. Using the results above, determine the matrix representation of the operator c acting on W with respect to the product basis.

2 Finite Group Representation Theory

Basic

Exercise 2.1. Let $G = S_3$ and let V be the standard 2-dimensional irreducible representation. The character values for V on the three conjugacy classes e , (12) , and (123) are given by:

$$\chi_V(e) = 2, \quad \chi_V((12)) = 0, \quad \chi_V((123)) = -1$$

Consider the tensor product representation $\rho = V \otimes V$.

- (a) Compute the character χ_ρ for all conjugacy classes and calculate the inner product $\langle \chi_\rho, \chi_\rho \rangle$.
- (b) Using the orthogonality relations, calculate the multiplicity of the trivial representation $\mathbf{1}$ in the decomposition of ρ . (Recall that $\chi_1(g) = 1$ for all g).

Intermediate

Exercise 2.2. Let $G = S_3$. Consider the set $X = \{(i, j) \mid 1 \leq i, j \leq 3, i \neq j\}$ of ordered pairs of distinct elements from $\{1, 2, 3\}$. The size of X is 6. S_3 acts on X naturally by $\sigma \cdot (i, j) = (\sigma(i), \sigma(j))$. Let π be the associated permutation representation on the vector space $\mathbb{C}[X]$.

- (a) Determine the character χ_π of this representation by counting the fixed points for representatives of each conjugacy class: e , (12) , and (123) .
- (b) Decompose χ_π into a sum of irreducible characters of S_3 .

Advanced

Exercise 2.3. Let $G = S_4$ and let W be the standard 3-dimensional irreducible representation defined by:

$$W = \left\{ (x_1, x_2, x_3, x_4) \in \mathbb{C}^4 \mid \sum_{i=1}^4 x_i = 0 \right\}$$

where G acts by permuting the coordinates.

Let $H = S_3$ be the subgroup of G consisting of permutations that fix the index 4 (i.e., permutations of $\{1, 2, 3\}$). Consider the permutation representation ℓ^H on the coset space $\mathbb{C}[G/H]$.

- (a) The multiplicity of an irreducible representation V_k in ℓ^H is given by $m_k = \dim(V_k^{*H})$, where V_k^{*H} is the subspace of vectors in V_k^* fixed by all elements of H .
Explicitly construct the subspace $W^H = \{w \in W \mid \sigma \cdot w = w, \forall \sigma \in H\}$.
- (b) Based on your construction in part (a), determine the dimension $\dim(W^H)$ and thus the multiplicity of the standard representation W in the decomposition of ℓ^H .

3 Solutions

3.1 Lie Algebra and Representation Theory

Solution to Problem 1

1. Proof that $\mathfrak{h}$ is a Lie subalgebra:

We calculate the Lie brackets for the basis elements d_{-1}, d_0, d_1 using the defining relation $[d_n, d_m] = (m - n)d_{n+m}$.

$$[d_0, d_{-1}] = (-1 - 0)d_{0+(-1)} = -d_{-1} \in \mathfrak{h}$$

$$[d_0, d_1] = (1 - 0)d_{0+1} = d_1 \in \mathfrak{h}$$

$$[d_1, d_{-1}] = (-1 - 1)d_{1+(-1)} = -2d_0 \in \mathfrak{h}$$

Since the result of the bracket operation between any two basis vectors lies within the span of the basis vectors $\{d_{-1}, d_0, d_1\}$, the subspace $\mathfrak{h}$ is closed under the Lie bracket. Thus, $\mathfrak{h}$ is a Lie subalgebra of W_1 .

2. Explicit Isomorphism:

Let $\phi : \mathfrak{sl}(2, \mathbb{C}) \rightarrow \mathfrak{h}$ be a linear map defined on the standard basis $\{e, f, h\}$ as follows:

$$\phi(e) = d_1, \quad \phi(f) = -d_{-1}, \quad \phi(h) = 2d_0.$$

We verify that this map preserves the Lie bracket relations of $\mathfrak{sl}(2, \mathbb{C})$:

• Relation $[h, e] = 2e$:

$$[\phi(h), \phi(e)] = [2d_0, d_1] = 2[d_0, d_1] = 2(d_1) = 2\phi(e). \quad \checkmark$$

- **Relation** $[h, f] = -2f$:

$$[\phi(h), \phi(f)] = [2d_0, -d_{-1}] = -2[d_0, d_{-1}] = -2(-d_{-1}) = 2d_{-1}.$$

On the other hand, $-2\phi(f) = -2(-d_{-1}) = 2d_{-1}$. Thus, the relation holds. ✓

- **Relation** $[e, f] = h$:

$$[\phi(e), \phi(f)] = [d_1, -d_{-1}] = -[d_1, d_{-1}] = -(-2d_0) = 2d_0 = \phi(h). \quad \checkmark$$

Thus, the map ϕ is an isomorphism of Lie algebras.

Solution to Problem 2

1. Rewriting x^2y^2 in terms of d :

We use the relation $xy = yx - 1 = d - 1$. First, we establish the commutation relation between x and d . Since $d = yx$:

$$dx = (yx)x = yx^2 = (xy + 1)x = x(yx) + x = xd + x = x(d + 1).$$

(Alternatively, using the commutator $[d, x] = y[x, x] + [y, x]x = 0 + 1 \cdot x = x$, so $dx - xd = x$).

Now we compute x^2y^2 :

$$\begin{aligned} x^2y^2 &= x(xy)y \\ &= x(d - 1)y \\ &= (xd)y - xy \\ &= (dx - x)y - (d - 1) \\ &= d(xy) - xy - d + 1 \\ &= d(d - 1) - (d - 1) - d + 1 \\ &= d^2 - d - d + 1 - d + 1 \\ &= d^2 - 3d + 2. \end{aligned}$$

Thus, the constants are $a = 1, b = -3, c = 2$.

2. Basis of A_1 :

A_1 is defined as the quotient of the free algebra $\mathbb{C}\langle x, y \rangle$ by the ideal $I = \langle yx - xy - 1 \rangle$. We

assume a filtration on A_1 where $\deg(x) = \deg(y) = 1$. The associated graded algebra $\text{gr}(A_1)$ is generated by $\bar{x}, \bar{y}$ with the relation:

$$\bar{y}\bar{x} - \bar{x}\bar{y} = 0$$

(since the non-commutative term 1 has degree 0, which is strictly lower than degree 2). Thus, $\text{gr}(A_1) \cong \mathbb{C}[x, y]$ (the commutative polynomial ring).

The set of monomials $\{x^i y^j \mid i, j \in \mathbb{N}\}$ forms a basis for the commutative polynomial ring $\mathbb{C}[x, y]$. By the standard argument for filtered algebras (analogous to the Poincaré-Birkhoff-Witt theorem), the set of ordered monomials $\{x^i y^j \mid i, j \in \mathbb{N}\}$ forms a vector space basis for A_1 .

Solution to Problem 3

1. Irreducibility of $V(-1, 0)$:

The action is given by $d_n \cdot v_k = (k - n)v_{k+n}$ (where we identify $v_k = z^k(dz)^{-1}$). This module is isomorphic to the adjoint representation of W_1 (via $v_k \leftrightarrow d_k$). Since W_1 is simple, its adjoint representation is irreducible.

Direct Proof: Let U be a non-zero submodule. Let $v = \sum a_k v_k \in U$ ($v \neq 0$). The operator d_0 acts diagonally ($d_0 v_k = k v_k$), so we can isolate individual basis vectors $v_k \in U$.

- Apply d_{-k} to v_k : $d_{-k} v_k = (k - (-k))v_0 = 2k v_0$. If $k \neq 0$, we generate v_0 .
- If we start with v_0 , applying d_n gives $d_n v_0 = (0 - n)v_n = -n v_n$. This generates all v_n for $n \neq 0$.
- If we start with v_{-1} , apply d_1 : $d_1 v_{-1} = (-1 - 1)v_0 = -2v_0$.

Thus, from any non-zero vector, we can generate the entire space. $V(-1, 0)$ is irreducible.

2. Reducibility of $V(1, 0)$:

The action is $d_n \cdot z^k dz = (k + n)z^{k+n} dz$. Let $v_k = z^k dz$. Consider the vector $v_0 = dz$ (index $k = 0$).

$$d_n \cdot v_0 = (0 + n)v_n = n v_n.$$

v_0 generates the whole module (for $n \neq 0$). However, can we reach v_0 from other vectors? Suppose we have v_k with $k \neq 0$. Applying d_{-k} :

$$d_{-k} \cdot v_k = (k + (-k))v_0 = 0 \cdot v_0 = 0.$$

The coefficient is always 0 when the target index is 0. Thus, v_0 cannot be generated by any other basis vector. The subspace spanned by all vectors *except* v_0 is invariant. Let $W = \text{span}_{\mathbb{C}}\{z^k dz \mid k \in \mathbb{Z}, k \neq 0\}$.

- If $k + n \neq 0$, $d_n v_k = (k + n)v_{k+n} \in W$.
- If $k + n = 0$, $d_n v_k = 0 \in W$.

Thus, W is a proper submodule, and $V(1, 0)$ is reducible.

3. Isomorphism of W :

Consider the module $V(0, 0) = \mathbb{C}[z, z^{-1}]$ with action $d_n z^k = k z^{k+n}$. The subspace $\mathbb{C} \cdot 1$ ($k = 0$) is a trivial submodule. The quotient $M = V(0, 0)/\mathbb{C}$ has a basis $\{[z^k] \mid k \neq 0\}$. The action on the quotient is $d_n [z^k] = k [z^{k+n}]$.

Define $\phi : W \rightarrow M$ by $\phi(z^k dz) = \frac{1}{k} [z^k]$ for $k \neq 0$. We check equivariance:

$$\phi(d_n \cdot z^k dz) = \phi((k + n)z^{k+n} dz) = (k + n) \frac{1}{k + n} [z^{k+n}] = [z^{k+n}].$$

$$d_n \cdot \phi(z^k dz) = d_n \left(\frac{1}{k} [z^k] \right) = \frac{1}{k} (k [z^{k+n}]) = [z^{k+n}].$$

The actions match, and ϕ is a bijection on the basis. Thus $W \simeq V(0, 0)/\mathbb{C}$.

Solution to Problem 4

1. Decomposition:

Using the Clebsch-Gordan rule for $\mathfrak{sl}(2, \mathbb{C})$:

$$V(3) \otimes V(1) \cong V(3 + 1) \oplus V(3 - 1) = V(4) \oplus V(2).$$

2. Highest Weight Vector of $V(2)$:

Let v_k denote a basis vector of weight $3 - 2k$ in $V(3)$ (where v_0 is HW) and u_k denote a basis vector of weight $1 - 2k$ in $V(1)$ (where u_0 is HW). The action of the lowering operator f (unnormalized) is $f v_k \propto v_{k+1}$. Specifically, for the highest weight vectors: $f v_0 = C_1 v_1$ and $f u_0 = C_2 u_1$. For simplicity, we assume standard normalization where coefficients are non-zero.

The highest weight vector of the $V(4)$ component is $w_{V(4)} = v_0 \otimes u_0$. The vector in $V(4)$

with weight 2 (one step down) is:

$$f(v_0 \otimes u_0) = (fv_0) \otimes u_0 + v_0 \otimes (fu_0) = 3v_1 \otimes u_0 + v_0 \otimes u_1.$$

(Note: Using the convention $f(x^k) = (m-k)x^{k+1}$ for $V(m)$, we get coeff 3 for $V(3)$ and 1 for $V(1)$).

The highest weight vector ψ of the $V(2)$ component must lie in the same weight space (weight $3+1-2=2$) and be orthogonal to $f(v_0 \otimes u_0)$ (or equivalently, annihilated by e). Let $\psi = A(v_1 \otimes u_0) + B(v_0 \otimes u_1)$. Applying the raising operator e (with $ev_1 = v_0$ and $eu_1 = u_0$, up to scaling):

$$\begin{aligned} e\psi &= A(ev_1 \otimes u_0 + v_1 \otimes eu_0) + B(ev_0 \otimes u_1 + v_0 \otimes eu_1) \\ &= A(v_0 \otimes u_0) + B(v_0 \otimes u_0) \\ &= (A+B)(v_0 \otimes u_0). \end{aligned}$$

For ψ to be a highest weight vector, we need $e\psi = 0$, which implies $A = -B$. Thus, the vector is (up to a scalar):

$$\psi = v_1 \otimes u_0 - v_0 \otimes u_1.$$

Solution to Problem 5

1. Decomposition of $W = V(1) \otimes V(1) \otimes V(1)$:

First step: $V(1) \otimes V(1) \cong V(2) \oplus V(0)$. Second step:

$$\begin{aligned} (V(2) \oplus V(0)) \otimes V(1) &\cong (V(2) \otimes V(1)) \oplus (V(0) \otimes V(1)) \\ &\cong (V(3) \oplus V(1)) \oplus V(1). \end{aligned}$$

Total decomposition:

$$W \cong V(3) \oplus V(1) \oplus V(1).$$

2. Highest Weight Vectors:

There are 3 highest weight vectors. Let e_1 (weight 1) and e_{-1} (weight -1) be the basis of $V(1)$. Notation: $e_{ijk} = e_i \otimes e_j \otimes e_k$.

- **For $V(3)$:** The unique state with max weight $1 + 1 + 1 = 3$.

$$w_{V(3)} = e_{111}$$

- **For the two $V(1)$ components:** These must have weight 1 and be annihilated by e .

First HW vector (via $V(2) \otimes V(1)$): The weight 0 state in $V(2)$ is proportional to $e_{1,-1} + e_{-1,1}$. Coupling $V(2)$ (weight 0) and $V(1)$ (weight 1) to get $V(1)$ requires the combination $v_{\text{mid}} \otimes u_{\text{high}} - v_{\text{high}} \otimes u_{\text{low}}$. Let $v_{\text{high}} = e_{11}$ and $v_{\text{mid}} = e_{1,-1} + e_{-1,1}$.

$$\psi_1 = (e_{1,-1} + e_{-1,1}) \otimes e_1 - 2(e_{11}) \otimes e_{-1} = e_{1,-1,1} + e_{-1,1,1} - 2e_{1,1,-1}.$$

Second HW vector (via $V(0) \otimes V(1)$): The $V(0)$ singlet state is $e_{1,-1} - e_{-1,1}$. Coupling this with $V(1)$ is trivial (tensor product).

$$\psi_2 = (e_{1,-1} - e_{-1,1}) \otimes e_1 = e_{1,-1,1} - e_{-1,1,1}.$$

3. Matrix of the Casimir Operator c :

The Casimir element acts as a scalar $(\lambda + 1)^2$ on $V(\lambda)$.

- On $V(3)$: eigenvalue $(3 + 1)^2 = 16$.
- On $V(1)$: eigenvalue $(1 + 1)^2 = 4$.

The matrix is block diagonal with respect to the weight spaces. We focus on the **weight 1 subspace**, spanned by $\{e_{1,1,-1}, e_{1,-1,1}, e_{-1,1,1}\}$. Let's denote these basis vectors as x, y, z respectively.

We know the spectral decomposition of the operator c on this subspace. One eigenvector is $v_3 = x + y + z$ (part of $V(3)$) with eigenvalue 16. Two eigenvectors (orthogonal complement) are in $V(1)$ with eigenvalue 4. The operator can be written as:

$$M = 16P_{V(3)} + 4P_{V(1) \oplus V(1)}.$$

Since $P_{V(1) \oplus V(1)} = I - P_{V(3)}$, we have $M = 4I + 12P_{V(3)}$. The projection onto the vector

$$\mathbf{1} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \text{ is } P_{V(3)} = \frac{1}{3}\mathbf{1}\mathbf{1}^T = \frac{1}{3} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}. \text{ Thus,}$$

$$M = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{pmatrix} + \begin{pmatrix} 4 & 4 & 4 \\ 4 & 4 & 4 \\ 4 & 4 & 4 \end{pmatrix} = \begin{pmatrix} 8 & 4 & 4 \\ 4 & 8 & 4 \\ 4 & 4 & 8 \end{pmatrix}.$$

The full matrix for W is block diagonal.

- On subspace $\{e_{111}\}$ (Weight 3): [16].
- On subspace $\{e_{1,1,-1}, e_{1,-1,1}, e_{-1,1,1}\}$ (Weight 1): $\begin{pmatrix} 8 & 4 & 4 \\ 4 & 8 & 4 \\ 4 & 4 & 8 \end{pmatrix}$.
- On subspace $\{e_{-1,-1,1}, e_{-1,1,-1}, e_{1,-1,-1}\}$ (Weight -1): $\begin{pmatrix} 8 & 4 & 4 \\ 4 & 8 & 4 \\ 4 & 4 & 8 \end{pmatrix}$.
- On subspace $\{e_{-1,-1,-1}\}$ (Weight -3): [16].

3.2 Finite Group Representation Theory

Solution to Problem 1

(a) Computing χ_ρ

According to Proposition 12.20, the character of a tensor product is the product of the characters:

$$\chi_\rho(g) = \chi_{V \otimes V}(g) = \chi_V(g) \cdot \chi_V(g) = (\chi_V(g))^2$$

We compute the values for each conjugacy class:

- Class e : $\chi_\rho(e) = (2)^2 = 4$.
- Class (12) : $\chi_\rho((12)) = (0)^2 = 0$.
- Class (123) : $\chi_\rho((123)) = (-1)^2 = 1$.

The sizes of the conjugacy classes in S_3 are: $|C_e| = 1$, $|C_{(12)}| = 3$, $|C_{(123)}| = 2$. The order of the group is $|G| = 6$.

Now, we calculate the inner product $\langle \chi_\rho, \chi_\rho \rangle$:

$$\begin{aligned}\langle \chi_\rho, \chi_\rho \rangle &= \frac{1}{|G|} \sum_{g \in G} |\chi_\rho(g)|^2 \\ &= \frac{1}{6} (1 \cdot (4)^2 + 3 \cdot (0)^2 + 2 \cdot (1)^2) \\ &= \frac{1}{6} (16 + 0 + 2) \\ &= \frac{18}{6} = 3.\end{aligned}$$

(b) Multiplicity of the Trivial Representation

Let n_1 be the multiplicity of the trivial representation $\mathbf{1}$ in ρ . By Corollary 14.14:

$$n_1 = \langle \chi_\rho, \chi_1 \rangle$$

Recall that $\chi_1(g) = 1$ for all g .

$$\begin{aligned}\langle \chi_\rho, \chi_1 \rangle &= \frac{1}{6} \sum_{\text{classes}} |C_g| \cdot \chi_\rho(g) \cdot \overline{\chi_1(g)} \\ &= \frac{1}{6} (1 \cdot 4 \cdot 1 + 3 \cdot 0 \cdot 1 + 2 \cdot 1 \cdot 1) \\ &= \frac{1}{6} (4 + 0 + 2) \\ &= \frac{6}{6} = 1.\end{aligned}$$

Thus, the multiplicity of the trivial representation is $\mathbf{1}$.

Solution to problem 2

(a) Determining the Character χ_π

For a permutation representation, the character value $\chi_\pi(g)$ is equal to the number of elements in the set X fixed by g .

$$\chi_\pi(g) = |\{x \in X \mid g \cdot x = x\}|$$

An element $(i, j) \in X$ is fixed by σ if and only if $\sigma(i) = i$ AND $\sigma(j) = j$.

- **Class e :** The identity fixes all elements. Since $|X| = 6$,

$$\chi_\pi(e) = 6.$$

- **Class (12) :** This permutation swaps $1 \leftrightarrow 2$ and fixes 3. For a pair (i, j) to be fixed, both i and j must be fixed points of (12) . The only fixed point is 3. However, elements in X must have distinct indices ($i \neq j$). We cannot form a pair $(3, 3)$ because i must not equal j . Thus, there are no pairs (i, j) with $i \neq j$ formed solely from the fixed point set $\{3\}$.

$$\chi_\pi((12)) = 0.$$

- **Class (123) :** This permutation cycles $1 \rightarrow 2 \rightarrow 3 \rightarrow 1$. It has no fixed points in $\{1, 2, 3\}$. Consequently, it cannot fix any pair (i, j) .

$$\chi_\pi((123)) = 0.$$

The character is $\chi_\pi = (6, 0, 0)$.

(b) Decomposition into Irreducible Characters

The irreducible characters of S_3 are:

- Trivial **1**: $(1, 1, 1)$
- Sign ε : $(1, -1, 1)$
- Standard V : $(2, 0, -1)$

We calculate the multiplicities $n_k = \langle \chi_\pi, \chi_k \rangle$:

1. Multiplicity of **1**:

$$n_1 = \frac{1}{6}(1 \cdot 6 \cdot 1 + 3 \cdot 0 \cdot 1 + 2 \cdot 0 \cdot 1) = \frac{6}{6} = 1.$$

2. Multiplicity of ε :

$$n_\varepsilon = \frac{1}{6}(1 \cdot 6 \cdot 1 + 3 \cdot 0 \cdot (-1) + 2 \cdot 0 \cdot 1) = \frac{6}{6} = 1.$$

3. Multiplicity of V :

$$n_V = \frac{1}{6}(1 \cdot 6 \cdot 2 + 3 \cdot 0 \cdot 0 + 2 \cdot 0 \cdot (-1)) = \frac{12}{6} = 2.$$

Check dimensions: $1 \cdot \dim(\mathbf{1}) + 1 \cdot \dim(\varepsilon) + 2 \cdot \dim(V) = 1 + 1 + 2(2) = 6 = \dim(\pi)$.

This matches.

Conclusion: The decomposition is:

$$\pi \cong \mathbf{1} \oplus \varepsilon \oplus 2V.$$

Solution to problem 3

(a) Construction of the Subspace W^H

Let $w \in W$. We can write w as a vector $(x_1, x_2, x_3, x_4) \in \mathbb{C}^4$. For w to be in W^H , it must satisfy two conditions:

1. **The Subspace Condition** ($w \in W$): The sum of coordinates must be zero.

$$x_1 + x_2 + x_3 + x_4 = 0 \quad (*).$$

2. **The Invariance Condition** (H -fixed): For all $\sigma \in H$, $\sigma \cdot w = w$. The subgroup $H = S_3$ permutes the indices $\{1, 2, 3\}$ and leaves index 4 fixed. Since H contains all transpositions of $\{1, 2, 3\}$, specifically (12) and (23), we must have:

- Invariance under (12): $(x_2, x_1, x_3, x_4) = (x_1, x_2, x_3, x_4) \implies x_1 = x_2$.
- Invariance under (23): $(x_1, x_3, x_2, x_4) = (x_1, x_2, x_3, x_4) \implies x_2 = x_3$.

Therefore, for w to be invariant under H , the first three coordinates must be equal. Let $x_1 = x_2 = x_3 = \alpha$ for some $\alpha \in \mathbb{C}$.

Now, substitute this into the subspace condition (*):

$$\alpha + \alpha + \alpha + x_4 = 0 \implies 3\alpha + x_4 = 0 \implies x_4 = -3\alpha.$$

Thus, any vector $w \in W^H$ must be of the form:

$$w = (\alpha, \alpha, \alpha, -3\alpha) = \alpha(1, 1, 1, -3).$$

This vector satisfies the sum condition ($1 + 1 + 1 - 3 = 0$), so it is indeed in W .

The subspace is explicitly:

$$W^H = \text{span}_{\mathbb{C}}\{(1, 1, 1, -3)\}.$$

(b) Dimension and Multiplicity

From the construction in part (a), we see that W^H is spanned by a single non-zero basis vector $v = (1, 1, 1, -3)$. Therefore:

$$\dim(W^H) = 1.$$

The multiplicity of the irreducible representation W in the decomposition of ℓ^H is equal to the dimension of the fixed-point subspace W^H .

Thus, the multiplicity of W in ℓ^H is **1**.